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1 Data Analysis Expressions (DAX)

Data Analysis Expression (DAX) is the formula language used in Microsoft Power BI, Power Pivot, and SQL Server Analysis Services (SSAS) Tabular Models. DAX enables analysts and developers to create custom calculations, business metrics, KPIs, and advanced analytical models.

DAX operates on columns, tables and relationships within a semantic model. It combines concepts from Excel formulas, relational databases, and functional programming to provide powerful analytical capabilities.

It allows users to:

- Create calculated columns
- Create measures
- Create calculated tables
- Perform aggregation.
- Perform filtering
- Perform ranking
- Perform statistical analysis
- Perform time intelligence calculations
- Build KPIs

Components of DAX : DAX consists of three major calculation objects:

- **Calculated Columns:** Calculated columns compute values row by row and become permanent columns within the data model. Examples: Profit
- **Measures:** Measures perform calculations dynamically based on filters applied in reports. Examples: Total Sales.
- **Calculated Tables:** Calculated tables generate entirely new tables using DAX expressions. Example: Calendar Table

Characteristics of DAX: DAX is:

- Column-based
- Relationship-aware
- Context-sensitive
- Case-insensitive
- Functional
- Optimized for analytical calculations
- Integrated with VertiPaq Engine

1.1 Create a measure to calculate the total sales amount from the sales table.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX formula:
 - Total Sales = SUM(Sales[SalesAmount])
- Press Enter
- Add the measure to a Card visual

1.2 Create a measure that calculates the total quantity sold.

Answer:

- Create a New Measure
- Enter the DAX
 - $\text{Total Quantity} = \text{SUM}(\text{Sales}[\text{Quantity}])$
- Press Enter
- Create a Card visual and drag: Total Quantity to Card

1.3 Create a calculated column that computes Profit for each sales transaction.

Answer:

- Select the Sales table
- Click New Column
- Enter the DAX expression
 - $\text{Profit} = \text{Sales}[\text{SalesAmount}] - \text{Sales}[\text{Cost}]$
- Press Enter

1.4 Create a measure to calculate the average sales amount.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - $\text{AverageSales} = \text{SUM}(\text{Sales}[\text{SalesAmount}])$
- Press Enter
- Add the measure to a Card visual

1.5 Create a measure that counts the total number of sales transaction.

Answer:

- In the Data/Fields pane on the right:
 - Find Sales
 - Click the Sales table
- Create a new measure
- Enter the DAX formula
 - $\text{TotalTransactions} = \text{COUNTROWS}(\text{Sales})$
- Create a Card visual
 - Go to Report view
 - Select Card
 - Drag Total Transactions into the Card

1.6 Create a measure that calculate the minimum sales amount recorded in the sales table.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `MinimumSales=MIN(Sales[SalesAmount])`
- Press Enter
- Add the measure to a Card visual

1.7 Create a measure that calculate the maximum sales amount.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `MaximumSales=MAX(Sales[SalesAmount])`
- Press Enter
- Add the measure to a Card visual

1.8 Create a measure to calculate the total profit.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `TotalProfit=SUM(Sales[Profit])`
- Press Enter
- Add the measure to a Card visual

1.9 Create a measure to calculate the average profit.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `AverageProfit=AVERAGE(Sales[Profit])`
- Press Enter
- Add the measure to a Card visual

1.10 Create a measure that counts the number of unique customers.

Answer:

- Open the Modeling tab
- Select New Measure
- Enter the DAX Formula
 - `TotalCustomers=DISTINCTCOUNT(Sales[CustomerID])`
- Press Enter
- Add the measure to a Card visual

1.11 Create a measure to calculate the total cost of all products sold.

Answer:

- Open the Modeling tab
- Select the Sales tables
- Click New Measure
- Enter the DAX formula
 - `TotalCost=SUM(Sales[Cost])`
- Press Enter
- Display the measure in a Card visual

1.12 Create a measure that calculate the total discount provided to customers.

Answer:

- Open the Modeling tab
- Select the sales tables
- Click New Measure
- Enter the DAX Formula
 - `TotalDiscount=SUM(Sales[Discount])`
- Press Enter
- Display the measure in a Card visual

1.13 Create a measure that counts the total number of products sold.

Answer:

- Open the Modeling tab
- Select the sales tables
- Click New Measure
- Enter the DAX Formula
 - `ProductsSold=COUNTRROWS(Sales)`
- Press Enter
- Display the measure in a Card visual

1.14 Create a measure to calculate the average quantity sold per transaction.

Answer:

- Open the Modeling tab
- Select the sales tables
- Click New Measure
- Enter the DAX Formula
 - `AverageQuantity=AVERAGE(Sales[Quantity])`
- Press Enter
- Display the measure in a Card visual

1.15 Create a measure that calculates the profit margin percentage.

Answer:

- Open the Modeling Tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - `ProfitMargin%=DIVIDE(SUM(Sales[Profit]),SUM(Sales[SalesAmount]),0)`
- Press Enter
- Display the measure in a card visual

1.16 Create a calculated column that classifies each sales as "High Value" or "Norma" based on the Sales Amount.If SalesAmount>=10000 then HighValue Otherwise Normal.

Answer:

- Open Power BI
- Select the sales Table
- Go to the Modeling tab
- Click New Column
- Enter the DAX Formula
 - `SalesCategory=IF(Sales[SalesAmount]>=10000,"HighValue","Normal")`
- Press Enter
- Display the column in a matrix visual

1.17 Create a calculated columns that combine First Name and Last Name into a Full Names.

Answer:

- Open Power BI
- Select the employees Table
- Go to the Modeling tab
- Click New Column
- Enter the DAX Formula
 - FullName=Employees[FirstName]&" "&Employees[LastName]
- Press Enter
- Display the column in a table visual

1.18 Create a measure to calculate Total Sales only for orders the Sales Amount is greater than Rs.50000.

Answer:

- Open the Modeling Tab
- Select the Employees Table
- Click New Measure
- Enter the DAX Formula
 - HighValueSales=CALCULATE(SUM(Sales[SalesAmount]),FILTER(Sales,Sales[SalesAmount]>50000))
- Press Enter
- Display a measure in a Card visual

1.19 Create a measure that calculate total Sales only for the "Electronic" product category.

Answer:

- Open the Modeling tab
- Select the sales Table
- Click New Measure
- Enter the DAX Formula
 - ElectronicSales=CALCULATE(SUM(Sales[SalesAmount]),Products[Category]="Electronics")
- Press Enter
- Display a measure in a card visual

1.20 Create a measure that returns the average sales amount for each customer.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New measure
- Enter the DAX Formula
 - `AverageSalesPerCustomer=DIVIDE(SUM(Sales[SalesAmount]),DISTINCTCOUNT(Sales[CustomerID]),0)`
- Press Enter
- Display a measure in a Card visual

1.21 Create a measure that calculate the percentage contribution of Total Profit to Total Sales.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New measure
- Enter the DAX Formula
 - `ProfitPercentage=DIVIDE(SUM(Sales[Profit]),SUM(Sales[SalesAmount]),0)`
- Press Enter
- Display a measure in a Card visual

1.22 Create a measure that calculate Year-to-Date(YTD) Sales.

Answer:

- Open the Modeling tab
- Select the sales table
- Ensure a proper Date table exists
- Mark it as the Date Table
- Create a new measure using TOTALYTD()
 - `YTDSales=TOTALYTD(SUM(Sales[SalesAmount]),'Date'[Date])`
- Use the Date column from the Date table

1.23 Create a measure that calculates Total Sales only for customers located in Delhi.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - `DelhiTotalSales=CALCULATE(SUM(Sales[SalesAmount]),Customers[City]="Delhi")`
- Press Enter
- Display a measure in a Card visual

1.24 Create a measure that returns the highest Sales amount.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - HighestSales=MAX(Sales[SalesAmount])
- Press Enter
- Display a measure in a Card visual

1.25 Create a measure that calculate Average Order Value.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - AverageOrderValue=DIVIDE(SUM(Sales[SalesAmount]),DISTINCTCOUNT(Sales[OrderID]),0)
- Press Enter
- Display a measure in a Card visual

1.26 Create a measure that returns the total sales for the current month.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - CurrentMonthSales = CALCULATE(SUM(Sales[SalesAmount]),FILTER(Sales,MONTH(Sales[OrderDate])=MONTH(TODAY())&& YEAR(Sales[OrderDate])=YEAR(TODAY())))
- Press Enter
- Display a measure in a Card visual

1.27 Create a measure that calculates Total Sales excluding cancelled orders.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - CompletedSales = CALCULATE(SUM(Sales[SalesAmount]),Sales[OrderStatus]<>"Cancelled")
- Press Enter
- Display a measure in a Card visual

1.28 Create a measure that calculates the percentage of completed orders.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - $\text{CompletedOrder\%} = \text{DIVIDE}(\text{CALCULATE}(\text{DISTINCTCOUNT}(\text{Sales}[\text{OrderID}]), \text{Sales}[\text{OrderStatus}] = \text{"Completed"}), \text{DISTINCTCOUNT}(\text{Sales}[\text{OrderID}]), 0)$
- Press Enter
- Display a measure in a Card visual

1.29 Create a measure that calculates Running Total Sales over time.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - $\text{RunningTotalSales} = \text{CALCULATE}(\text{SUM}(\text{Sales}[\text{SalesAmount}]), \text{FILTER}(\text{ALL}('Date'[\text{Date}]), 'Date'[\text{Date}] \leq \text{MAX}('Date'[\text{Date}])))$
- Press Enter
- Display a measure in a Card visual

1.30 Create a measure that calculate Year-over-Year (YoY) Sales Growth Percentage.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - $\text{YoYGrowth\%} = \text{VAR CurrentSales} = \text{SUM}(\text{Sales}[\text{SalesAmount}]) \text{ VAR PreviousSales} = \text{CALCULATE}(\text{SUM}(\text{Sales}[\text{SalesAmount}]), \text{SAMEPERIODLASTYEAR}('Date'[\text{Date}])) \text{ RETURN DIVIDE}(\text{CurrentSales} - \text{PreviousSales}, \text{PreviousSales}, 0)$
- Press Enter
- Display a measure in a Card visual

1.31 Create a measure that returns the Top 5 Customers based on Total Sales.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - `Top5CustomerSales = SUMX(TOPN(5,SUMMARIZE(Sales,Sales[CustomerID], "Sales",SUM(Sales[SalesAmount])),[Sales],DESC),[Sales])`
- Press Enter
- Display a measure in a Card visual

1.32 Create a measure that ranks products according to Total Sales.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - `ProductRank = RANKX(ALL(Products[ProductName]),[Total Sales],,DESC,Dense)`
- Press Enter
- Display a measure in a Card visual

1.33 Create a dynamic KPI that displays either Total Sales, Total Profit, or Total Quantity depending on the user's slicer selection.

Answer:

- Open the Home tab
- Click Enter data
- Create a table named Metric
- Enter the following values:
 - Metrics
 - Sales
 - Profit
 - Quantity
- Click Load
- Select the Metric table
- Do not create any relationship with the sales table
- Open the Modeling tab
- Click New Measure
- Enter the DAX Formula:

- SelectedKPI = SWITCH(SELECTEDVALUE('Metric (2)'[Metric]),"Sales",[Total Sales],
"Profit",[TotalProfit],"Quantity",[Total Quantity],BLANK())

- Press Enter
- Insert a Slicer visual
- Drag Metric[Metric] into the Slicer
- Enable Single Select for the slicer
- Insert a Card visual
- Drag Selected KPI into the Card
- Select Sales, Profit, or Quantity from the slicer
- The Card displays the corresponding KPI dynamically

1.34 Create a measure that calculates Month-over-Month (MoM) Sales Growth Percentage.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - MoM Growth% = VAR CurrentSales=[Total Sales] VAR PreviousMonthSales=
CALCULATE([Total Sales],DATEADD('Date'[Date],-1,MONTH))
RETURN DIVIDE(CurrentSales-PreviousMonthSales,PreviousMonthSales,0)
- Press Enter
- Display a measure in a Card visual

1.35 Create a measure that displays the sales contribution percentage of each customer relative to overall company sales.

Answer:

- Open the Modeling tab
- Select the sales table
- Click New Measure
- Enter the DAX Formula
 - CustomerContribution% = DIVIDE([Total Sales],CALCULATE([Total Sales],ALL(Customers)),0)
- Press Enter
- Display a measure in a Card visual

1.36 Create a measure that returns a Dynamic Sales Target Status based on whether Total Sales achieve a target of Rs.50,00,000.

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - SalesTargetStatus = IF([Total Sales]>=5000000,"Target Achieved","Target Pending")
- Press Enter
- Display a measure in a Card Visual

1.37 Create a measure that calculates the Moving Average of Sales over the last 3 months.

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - ThreeMonthMovingAverage = AVERAGEX(DATESINPERIOD('Date'[Date],MAX('Date'[Date]),-3,MONTH),[Total Sales])
- Press Enter
- Display a measure in a Card Visual

1.38 Create a measure that ranks customers based on Total Sales while respecting report filters.

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - CustomerRank = RANKX(ALLSELECTED(Customers[CustomerName]),[Total Sales],,DESC,Dense)
- Press Enter
- Display a measure in a Card visual

1.39 Create a measure that returns Sales only for the Top 20% of customers based on TotalSales.

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - Top20PercentSales = VAR CustomerCount=COUNTROWS(VALUES(Customers[CustomerID]))
VAR TopCustomers=TOPN(ROUNDUP(CustomerCount*0.20,0),
SUMMARIZE(Customers,Customers[CustomerID],"Sales",[Total Sales]),[Sales],DESC)
RETURN SUMX(TopCustomers,[Sales])
- Press Enter
- Display a measure in a Card visual

1.40 Create a measure that calculates the percentage of TotalSales contributed by the currently selected region.

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - `CustomerContribution% = DIVIDE([Total Sales],CALCULATE([Total Sales],ALL(Customers)),0)`
- Press Enter
- Display a measure in a Card Visual

1.41 Create a dynamic measure that displays Sales, Profit, Quantity, or Orders based on slicer selection.

Answer:

- Go to the Home tab
- Click Enter data
- A table window will appear.
- Rename the first column to:
 - Metric
- Enter the following values:
 - Metric
 - Sales
 - Profit
 - Quantity
- At the bottom, change the table name to: Metric
- Click Load
- Do not create a relationship
- Create Total Sales Measure
- Create Total Profit Measure
- Create Total Quantity measure
- Create the dynamic KPI measure
- Enter the following DAX:
 - `SelectedKPI = SWITCH(SELECTEDVALUE('Metric (2)'[Metric]),"Sales",[Total Sales], "Profit",[TotalProfit],"Quantity",[Total Quantity],BLANK())`
- Create the slicer
 - Go to Report view
 - Click an empty area of the report canvas
 - In the Visualizations pane, select Slicer
 - From the Data pane, find: Metric
 - Drag: Metric[Metic] into the slicer.

- Enable Single Select
 - Select the Format pane
 - Find the Selection settings
 - Turn Single select to On
- Insert a Card visual with Selected KPI

1.42 Create a measure that calculates Sales for the last 12 months regardless of the reports' current date filter.

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - Rolling12MonthSales = CALCULATE([Total Sales],DATESINPERIOD('Date'[Date],MAX('Date'[Date]),-12,MONTH))
- Press Enter
- Display a measure in a Card Visual

1.43 Create a measure that returns the cumulative sales percentage (Pareto Analysis).

Answer:

- Open the Modeling tab
- Select the sales table
- Enter the DAX Formula:
 - ParetoSales% = VAR CurrentRank=RANKX(ALL(Customers[CustomerName]),[Total Sales],,DESC)

VAR RunningSales=SUMX(FILTER(ALL(Customers[CustomerName]),

RANKX(ALL(Customers[CustomerName]),[Total Sales],,DESC)<=CurrentRank),[Total Sales])

RETURN DIVIDE(RunningSales,CALCULATE([Total Sales],ALL(Customers)),0)
- Press Enter
- Display a measure in a Card Visual

1.44 Create a dynamic measure that displays Sales, profit, Quantity,or Orders based on slicer selection.

Answer:

- Go to the Home tab
- Click Enter data
- A table window will appear
- Rename the first column to Metric
- Enter the following values:
 - Metric
 - Sales
 - Profit

- Quantity
 - Orders
- At the bottom, change the table name to: Metric Selector
- Click Load
- Do not create a relationship
- Create Total Sales Measure
 - TotalSales=SUM(Sales[SalesAmount])
- Create Total Profit Measure
 - TotalProfit=SUM(Sales[Profit])
- Create Total Quantity Measure
 - TotalQuantity=SUM(Sale[Quantity])
- Create Total Orders Measure
 - TotalOrders=DISTINCTCOUNT(Sales[OrderID])
- Create the dynamic KPI measure
- Enter the following DAX:
 - ExecutiveKPI = VAR SelectedMetric = SELECTEDVALUE('MetricSelector'[Metric],"Sales")
RETURN SWITCH(SelectedMetric,"Sales",[Total Sales],"Profit",[TotalProfit]
,"Quantity",[Total Quantity],"Orders",[TotalOrders],"Customers",[TotalCustomers],BLANK())
- Create the slicer
 - Go to Report view
 - Click an empty area of the report canvas
 - In the Visualizations pane, select Slicer
 - Drag MetricSelector[Metric] into the slicer
- Enable Single Select for the slicer
- Create a Card visual
- Drag ExecutiveKPI into the Card